

BREATHEBEAT - Individual Sprint 3 Report

Literature Review and Audio Design Contribution

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1. Motivation for the Sprint 3 Redesign

In Sprint 2, our prototype mapped breathing directly to a melodic traversal of Raga Bhupali: 4 s inhalation, 2 s hold, and 6 s exhalation, supported by a Tanpura drone.

Upon testing the manually constructed prototype, we found that the audio was not suitable for a cognitive recovery intervention. The discrete pitch changes were too perceptible, transitions sounded abrupt, and the repeated scale traversal made the breathing cue feel like a complex sequence that required active cognitive effort to follow.

Consequently, Sprint 3 focuses on grounding our audio design in established experimental literature. The goal is to identify restorative acoustic features, refine our testing approach, and finalize a non-distracting intervention before running our main participant cohort.

2. Literature Review

2.1 Auditory Breathing Pacing (Gummidela et al., 2021)

Gummidela et al. developed an auditory breathing-guidance intervention that avoided melodies and spoken counts. Instead, it used a **continuous amplitude envelope**: sound intensity gradually increased during inhalation and decreased during exhalation [1]. This pacing condition produced a high proportion of immediate self-reported relaxation. For our hypothesis, this provides a direct method for converting respiration into a smooth acoustic envelope rather than relying on discrete musical notes.

2.2 Purpose-Built Instrumental Music (Burns et al., 2013)

Burns, Drews, and Carpenter developed an audio-based paced-respiration intervention using **instrumental music coupled with an audible breathing cue**. The music was explicitly composed for the intervention, embedding the breathing cue into the musical structure rather than forcing a breathing cycle onto an existing song [2]. This strongly supports our approach of purpose-building the BREATHEBEAT audio engine rather than using standard Indian Classical recordings.

2.3 Indian Classical Audio: Bhupali and Tanpura

Nagarajan et al. demonstrated that listening to instrumental Raga Bhupali significantly improved forward and backward digit-span scores in healthy students, while also reducing anxiety [3, 4]. Furthermore, Ubrangala et al. used tailor-made classical recordings consisting of a **Tanpura drone and Bansuri alaap** (without rhythmic percussion or fixed tempo) to produce sustained parasympathetic responses [5]. These studies validate the use of Bhupali and Tanpura as our foundational acoustic environment.

3. Audio Testing and Feature Selection

To ensure the final BREATHEBEAT audio is restorative rather than distracting, we will conduct a preliminary feature-evaluation phase (a micro-pilot study) before exposing the main cohort of participants to the system.

Instead of arbitrarily selecting an audio format, we will test prototypes internally to evaluate how specific musical features impact subjective cognitive load. The testing will isolate and evaluate the following components:

- **Tempo and Pacing:** Determining whether a continuous, equal-duration breathing rhythm (e.g., 5 s inhale, 5 s exhale) feels more natural and less forced than extended exhalations.
- **Melodic Movement:** Comparing discrete note triggers against continuous pitch glides (*meend*) to identify which approach requires less active listening.
- **Acoustic Timbre:** Testing the difference between raw synthetic oscillators and natural instrument sound profiles (e.g., Bansuri flute) to maximize emotional resonance and comfort.

References

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